

1) List the transformations of, in words, of $f(x)$ for each of the following functions.

a) $g(x) = 2f(x)$

- vertical stretch by a factor of 2

b) $h(x) = f(x - 3)$

- phase shift right 3 units

c) $j(x) = f\left(\frac{1}{3}x\right)$

- horizontal stretch by a factor of 3

d) $k(x) = f(-x)$

- horizontal reflection

e) $m(x) = f(x) - 3$

- shift down 3 units.

2) List the transformations, in words, of $f(x)$ for each of the following functions in the order you would do them in.

a) $g(x) = -f(x + 2)$

- vertical reflection

- phase shift left 2 units

b) $h(x) = f(3x) + 2$

- horizontal compression by a factor of $\frac{1}{3}$

- shift up 2 units

c) $j(x) = 3f(-x)$

- vertical stretch by a factor of 3

- horizontal reflection

3) List the transformations, in words, of $f(x) = x^2$ to $g(x) = 3(x - 2)^2 - 11$ in the order you would do them.

- vertical stretch by a factor of 3

- phase shift right 2 units

- shift down 11 units

4) List the transformations, in words of $f(x) = \sqrt{x}$ to $g(x) = 2\sqrt{(x - 3)} - 9$ in the order you would do them.

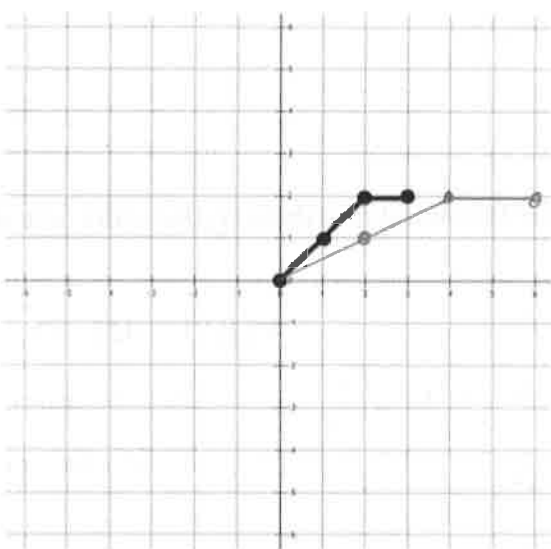
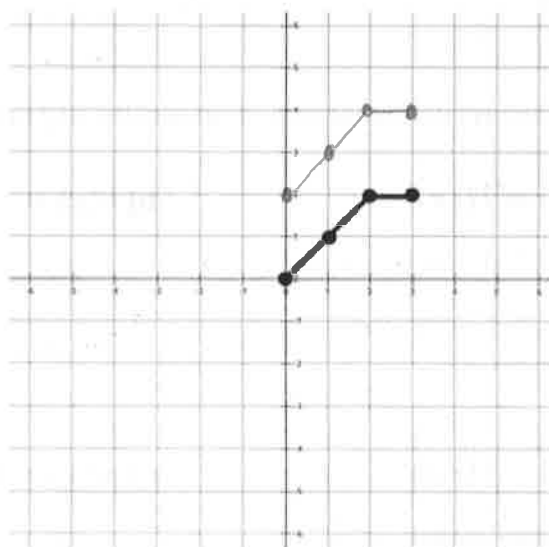
- vertical stretch by a factor of 2

- phase shift right 3 units

- shift down 9 units

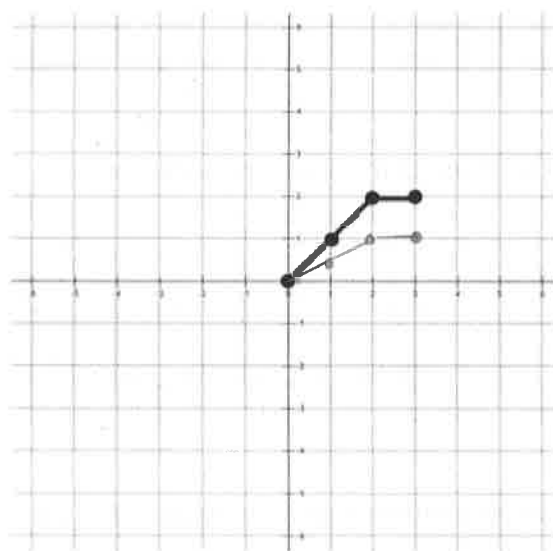
5) Perform the following transformations on the graphs below.

a) translate up 2 units



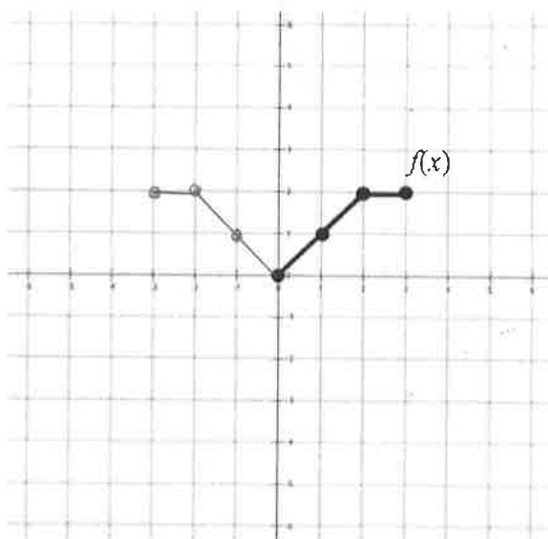
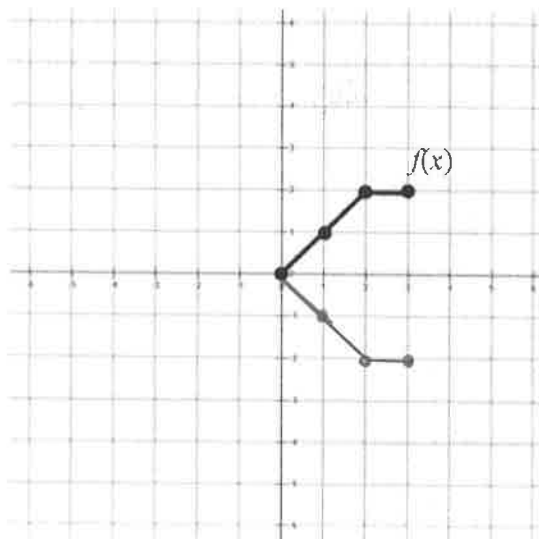
b) horizontal stretch by a factor of 2

c) vertical ^{compression} stretch by a factor of $\frac{1}{2}$



6) For the graph of $f(x)$ given, sketch the graph of $g(x)$ after the given transformation. List the transformations in words as well.

a) $g(x) = -f(x)$



b) $g(x) = f(-x)$

7) For each function $g(x)$:

- describe the transformations from the parent function $f(x)$
- create a table of values of image points for the transformed function
- graph the parent function and the transformed function and write its equation

a) $f(x) = x^2$. Graph $g(x) = 2f(x - 2)$.

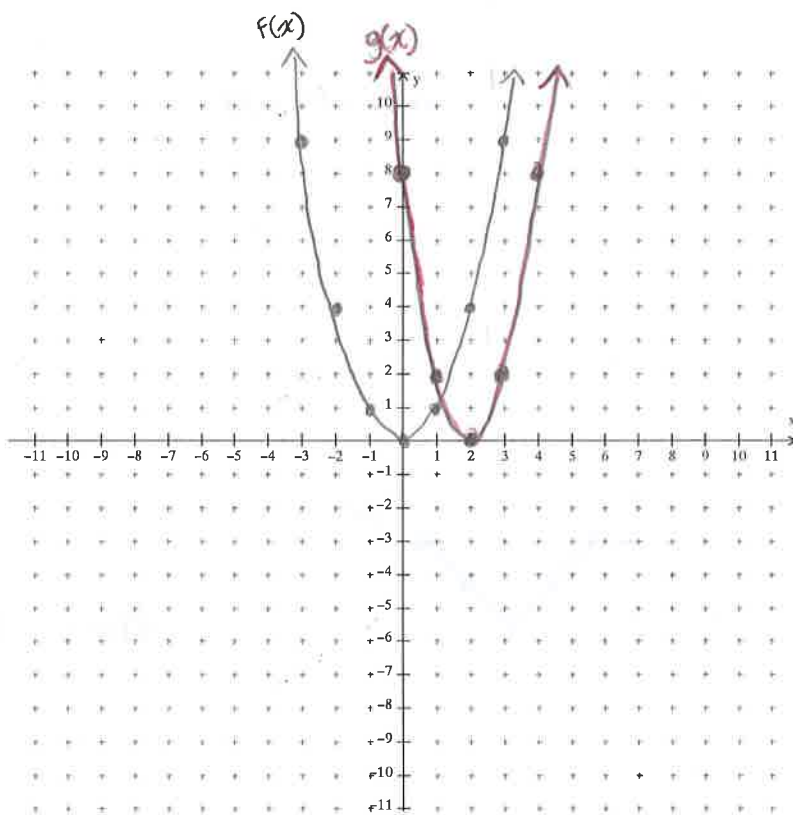
- vertical stretch BAFO 2 (2y)
shift right 2 units ($x+2$)

ii)

$f(x)$	$g(x)$	
	$x+2$	$2y$
$(-3, 9)$	-1	18
$(-2, 4)$	0	8
$(-1, 1)$	1	2
$(0, 0)$	2	0
$(1, 1)$	3	2
$(2, 4)$	4	8
$(3, 9)$	5	18

$$f(x) = x^2$$

$$g(x) = 2(x-2)^2$$



b) $f(x) = \sqrt{x}$. Graph $g(x) = -f(2x)$.

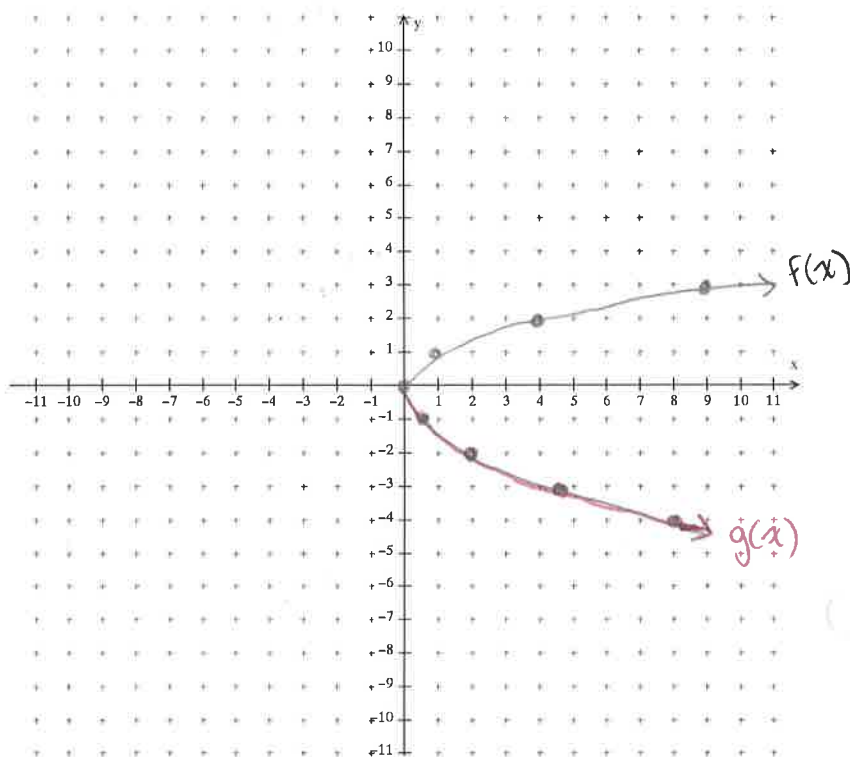
- vertical reflection ($-y$)
horizontal compression by $\frac{1}{2}$ ($\frac{x}{2}$)

ii)

$f(x)$	$g(x)$	
	$\frac{x}{2}$	$-y$
$(0, 0)$	0	0
$(1, 1)$	0.5	-1
$(4, 2)$	2	-2
$(9, 3)$	4.5	-3

$$f(x) = \sqrt{x}$$

$$g(x) = -\sqrt{2x}$$



c) $f(x) = x^2$. Graph $g(x) = 4f(x - 3)$

i) vertical stretch BAFO 4 (4y)
shift right 3 units ($x+3$)

ii)

$f(x)$

$(-3, 9)$

$(-2, 4)$

$(-1, 1)$

$(0, 0)$

$(1, 1)$

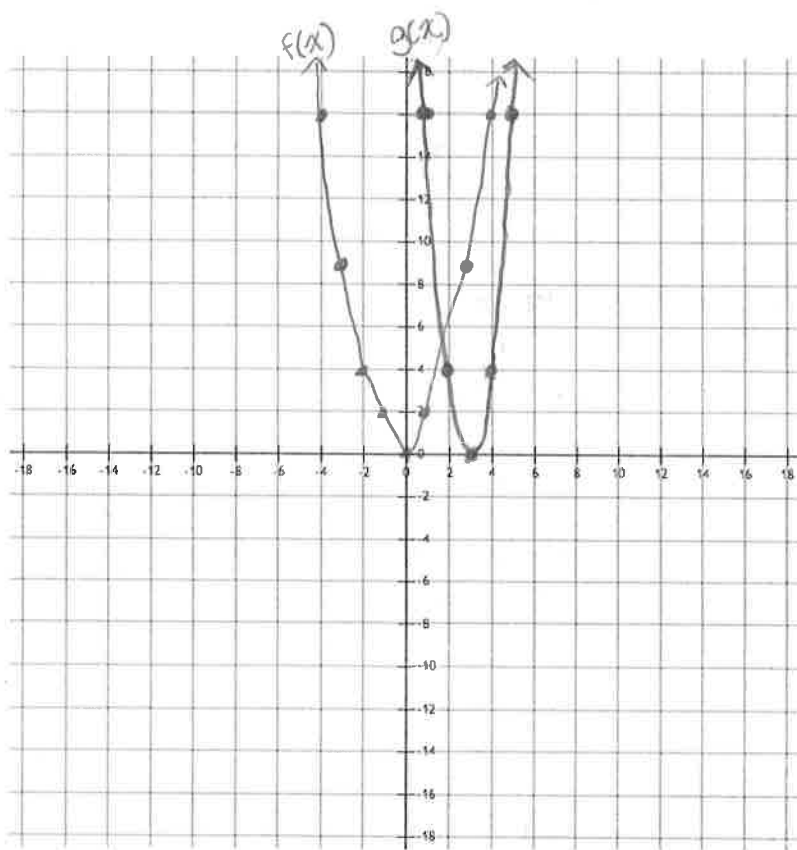
$(2, 4)$

$(3, 9)$

$x+3$	$4y$
0	36
1	16
2	4
3	0
4	4
5	16
6	36

$$f(x) = x^2$$

$$g(x) = 4(x-3)^2$$



d) $f(x) = \sqrt{x}$. Graph $g(x) = 3f(-x) - 2$.

i) vertical stretch BAFO 3 (3y)

horizontal reflection ($-x$)

shift down 2 units ($y-2$)

ii)

$f(x)$

$(0, 0)$

$(1, 1)$

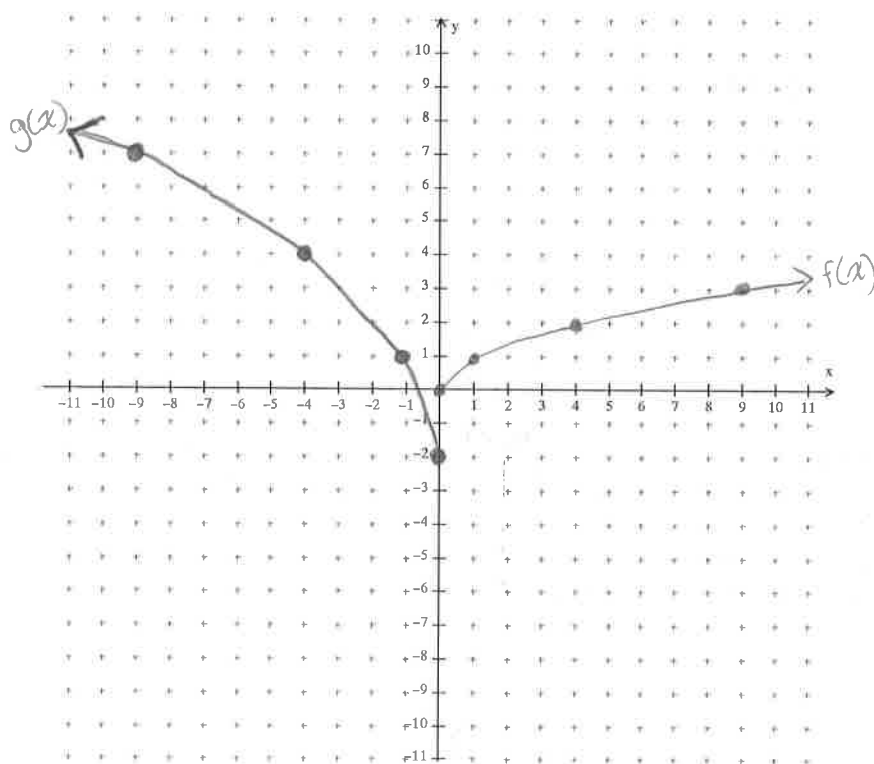
$(4, 2)$

$(9, 3)$

$-x$	$3y-2$
0	-2
-1	1
-4	4
-9	7

$$f(x) = \sqrt{x}$$

$$g(x) = 3\sqrt{-x} - 2$$



only graph $g(x)$

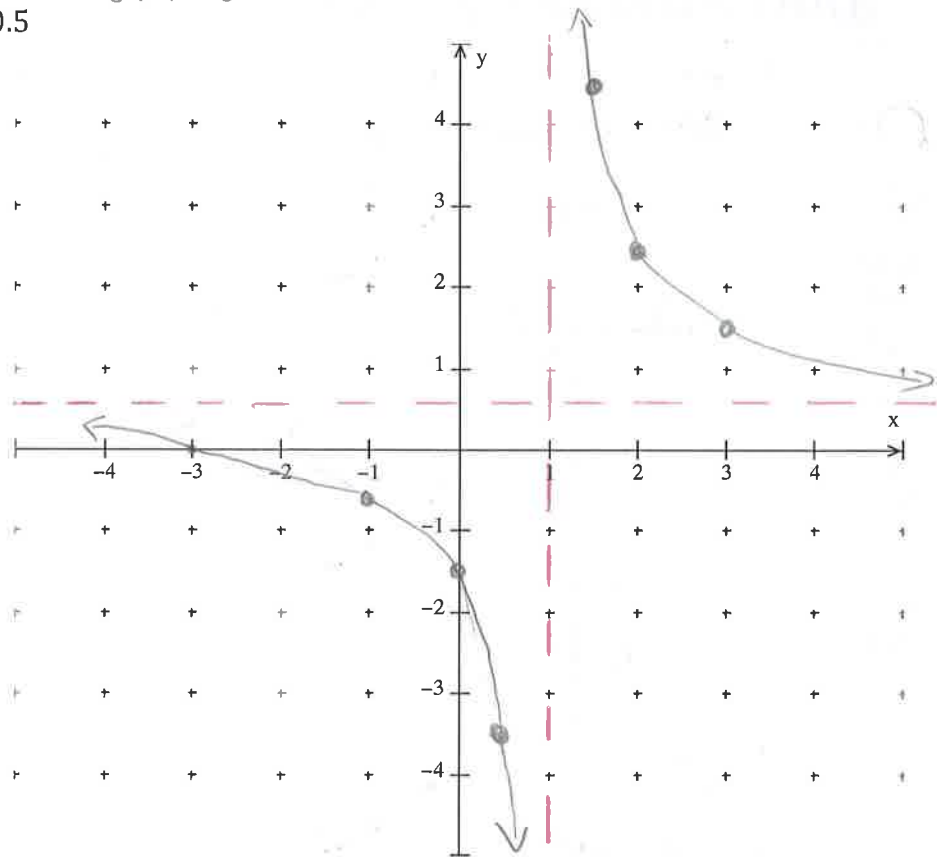
e) $f(x) = \frac{1}{x}$. Graph $g(x) = 2f(x-1) + 0.5$

i) vertical stretch base 2 ($2y$)

shift 1 unit right ($x+1$)

shift up 0.5 units ($y+0.5$)

$f(x)$	$g(x)$
$(-2, -0.5)$	$x+1$ $2y+0.5$
$(-1, -1)$	-1 -0.5
$(-0.5, -2)$	0 -1.5
$(0.5, 2)$	0.5 -3.5
$(1, 1)$	1.5 4.5
$(2, 0.5)$	2 2.5
	3 1.5
$g(x) = \frac{2}{x-1} + 0.5$	



8) For each function $g(x)$:

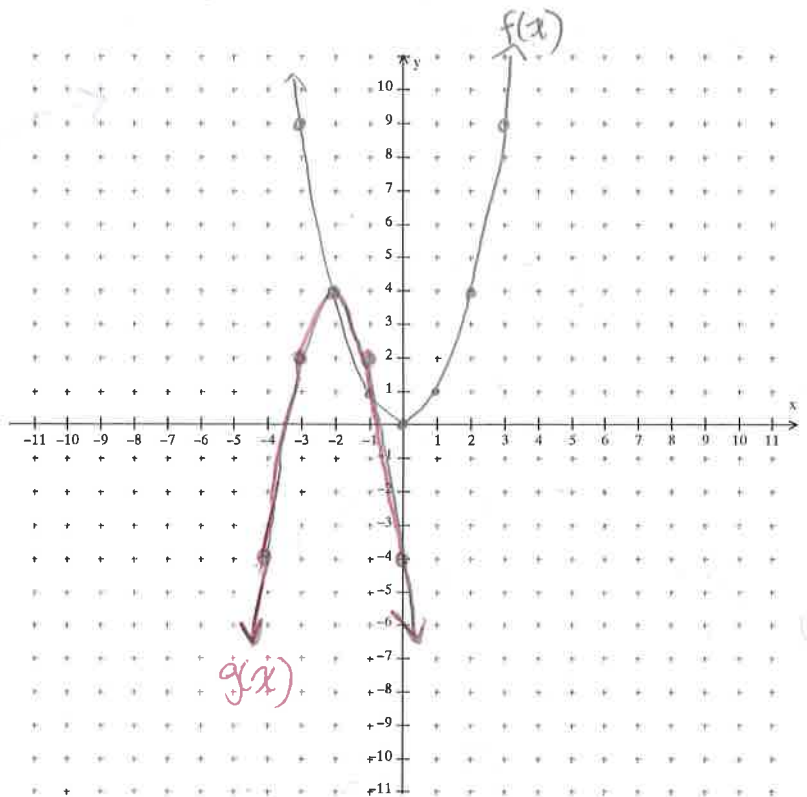
- determine the parent function and describe the transformations from the parent function $f(x)$
- create a table of values of image points for the transformed function
- graph the parent function and the transformed function

a) $g(x) = -2(x+2)^2 + 4$

$f(x) = x^2$

Transformations: vertical stretch base 2 ($2y$)
vertical reflection ($-y$)
left 2 units ($x-2$)
up 4 units ($y+4$)

$f(x)$	$g(x)$
$(-3, 9)$	$x-2$ $-2y+4$
$(-2, 4)$	-5 -14
$(-1, 1)$	-4 -4
$(0, 0)$	-3 2
$(1, 1)$	-2 4
$(2, 4)$	-1 2
$(3, 9)$	0 -4
	1 -14



$$f(x) = \sqrt{x}$$

b) $g(x) = -2\sqrt{-4(x+2)} - 3$

i) vertical stretch by 2 ($2y$)

vertical reflection ($-y$)

horizontal compression by $\frac{1}{4}$ ($\frac{x}{4}$)

horizontal reflection ($-x$)

shift left 2 ($x-2$)

down 3 ($y-3$)

$f(x)$

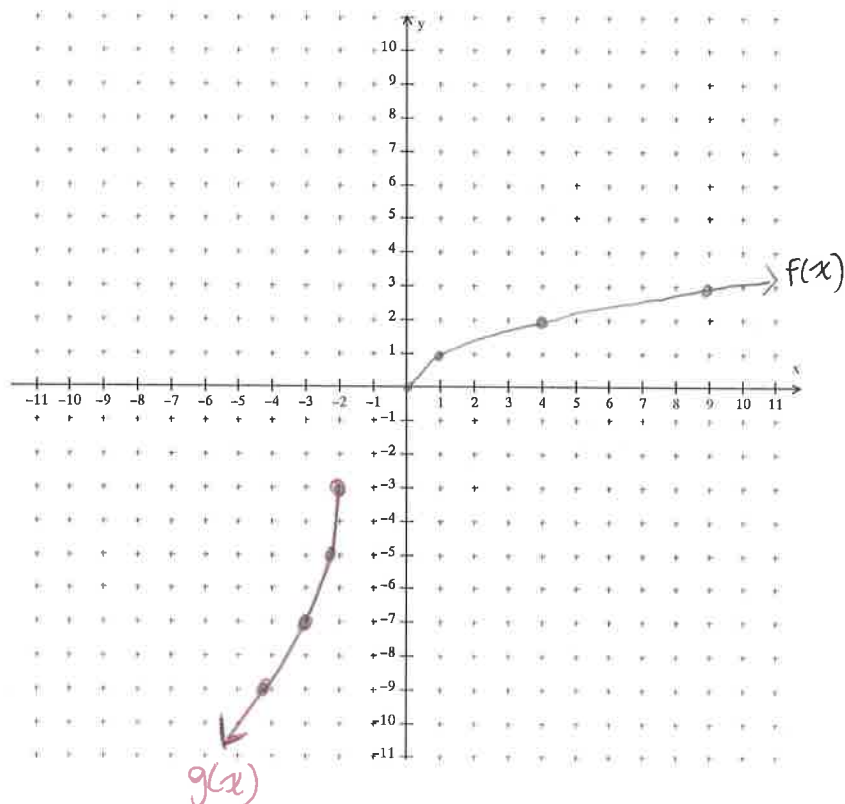
(0,0)

(1,1)

(4,2)

(9,3)

$\frac{x}{4} - 2$	$-2y - 3$
-2	-3
-2.25	-5
-3	-7
-4.25	-9



$$f(x) = \frac{1}{x}$$

c) $g(x) = \frac{-1}{\frac{1}{2}(x+0.5)} + 1$

only graph $g(x)$

i) vertical reflection ($-y$)

horizontal stretch by 2 ($2x$)

left 0.5 units ($x-0.5$)

up 1 unit ($y+1$)

$f(x)$

(-2, -0.5)

(-1, -1)

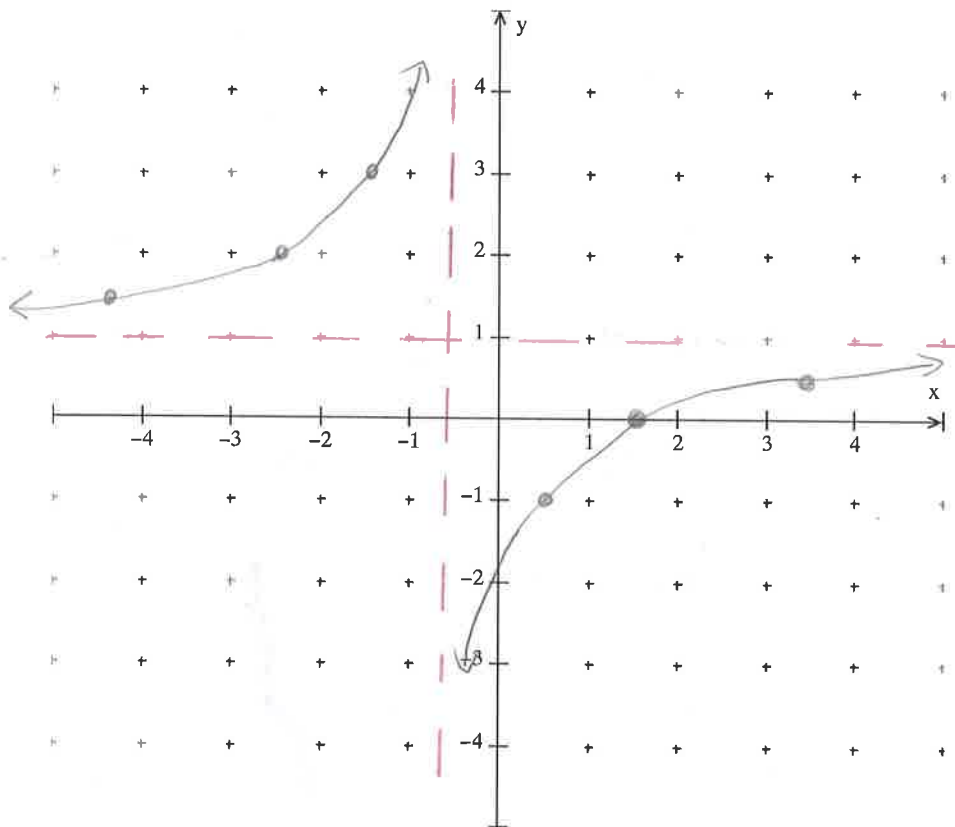
(-0.5, -2)

(0.5, 2)

(1, 1)

(2, 0.5)

$2x - 0.5$	$-y + 1$
-4.5	1.5
-2.5	2
-1.5	3
0.5	-1
1.5	0
3.5	0.5



9) Find the inverse, $f^{-1}(x)$, algebraically if $f(x) = -2\sqrt{x+1} - 5$

$$x = -2\sqrt{y+1} - 5$$

$$\frac{x+5}{-2} = \sqrt{y+1}$$

$$\left(\frac{x+5}{-2}\right)^2 = y+1$$

$$\left(\frac{x+5}{-2}\right)^2 - 1 = y$$

$$f^{-1}(x) = \left(\frac{x+5}{-2}\right)^2 - 1$$

10) Find the inverse, $f^{-1}(x)$, algebraically if $f(x) = \frac{1}{3}(x-4)^2 + 2$

$$x = \frac{1}{3}(y-4)^2 + 2$$

$$x-2 = \frac{1}{3}(y-4)^2$$

$$3x-6 = (y-4)^2$$

$$\pm\sqrt{3x-6} = y-4$$

$$\pm\sqrt{3x-6} + 4 = y$$

$$f^{-1}(x) = \pm\sqrt{3x-6} + 4$$

OR

$$x = \frac{1}{3}(y-4)^2 + 2$$

$$3(x-2) = (y-4)^2$$

$$\pm\sqrt{3(x-2)} = y-4$$

$$\pm\sqrt{3(x-2)} + 4 = y$$

$$f^{-1}(x) = \pm\sqrt{3(x-2)} + 4$$

Answers

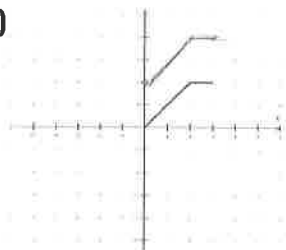
- 1) a) vertical stretch BAFO 2 b) phase shift right 3 units c) horizontal stretch BAFO 3
d) horizontal reflection in the y-axis e) shift down 3 units

- 2) a) vertical reflection in the x-axis and then shift left 3 units.
b) horizontal compression BAFO $\frac{1}{3}$ and then shift up 2 units.
c) vertical stretch BAFO 3 and then horizontal reflection in the y-axis.

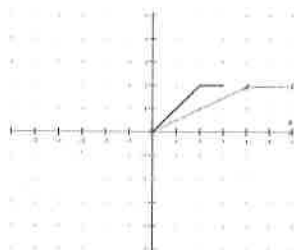
- 3) vertical stretch BAFO 3, then shift right 2 units and down 11 units.

- 4) vertical stretch by a factor of 2, then shift right 2 units and down 9 units.

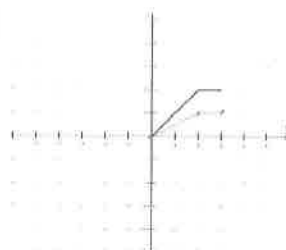
5) a)



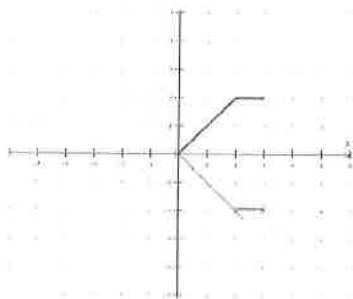
b)



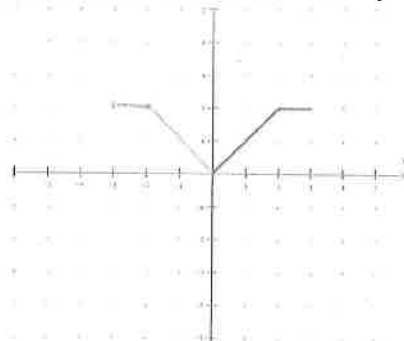
c)



- 6) a) vertical reflection in the x-axis



- b) horizontal reflection in the y-axis



See posted solutions for 7&8

9) $f^{-1}(x) = \left(\frac{x+5}{-2}\right)^2 - 1$

10) $f^{-1}(x) = \pm\sqrt{3(x-2)} + 4$

